

第十八讲：协方差与相关系数

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定义 0.1 对于随机向量 $X = (X_1, \dots, X_n)$, 类似于随机向量期望的定义, 我们定义随机向量 X 的方差为 $D(X) := (D(X_1), D(X_2), \dots, D(X_n))$.

定义 0.2 设随机向量 $(X_1, \dots, X_n)$ 的每个分量 X_i 的方差均存在, 称

$$\text{cov}(X_i, X_j) := E[(X_i - EX_i)(X_j - EX_j)], i, j = 1, 2, \dots, n$$

为 X_i 与 X_j 的协方差. 而将协方差构成的 $n \times n$ 方阵

$$B = (b_{ij}), \quad b_{ij} = \text{cov}(X_i, X_j)$$

称为随机向量的协方差阵。一般来说,

- ▶ 若 $\text{cov}(X_i, X_j) > 0$ 时, 称 X_i 与 X_j 正相关;
- ▶ 若 $\text{cov}(X_i, X_j) < 0$ 时, 称 X_i 与 X_j 负相关;
- ▶ 若 $\text{cov}(X_i, X_j) = 0$ 时, 称 X_i 与 X_j 不相关或零相关.

协方差的性质: $cov(X_i, X_j) = E[(X_i - EX_i)(X_j - EX_j)]$



- ▶ $cov(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$: 事实上

$$\begin{aligned} cov(X_i, X_j) &= E[(X_i - EX_i)(X_j - EX_j)] \\ &= E\left[X_i X_j - X_i E(X_j) - X_j E(X_i) + E(X_i)E(X_j)\right] \\ &= E(X_i X_j) - E(X_i)E(X_j) \end{aligned}$$

- ▶ 若 a 为常数, 则 $cov(X_i, a) = 0$;
- ▶ 对任意常数 a, b , 有 $cov(aX_i, bX_j) = ab \cdot cov(X_i, X_j)$;
- ▶ $cov(X_i + X_j, X_k) = cov(X_i, X_k) + cov(X_j, X_k)$.

协方差与方差: $cov(X_i, X_j) = E[(X_i - EX_i)(X_j - EX_j)]$



- ▶ $cov(X_i, X_i) = E(X_i - EX_i)^2 = D(X_i)$;
- ▶ $D(X_1 + X_2 + \cdots + X_n) = \sum_{i=1}^n D(X_i) + 2 \sum_{1 \leq i < j \leq n} cov(X_i, X_j)$: 事实上

$$\begin{aligned} D\left(\sum_{i=1}^n X_i\right) &= E\left(\sum_{i=1}^n X_i - E\sum_{i=1}^n X_i\right)^2 = E\left(\sum_{i=1}^n (X_i - EX_i)\right)^2 \\ &= E\left[\sum_{i=1}^n (X_i - EX_i)^2 + 2 \sum_{1 \leq i < j \leq n} (X_i - EX_i)(X_j - EX_j)\right] \\ &= \sum_{i=1}^n D(X_i) + 2 \sum_{1 \leq i < j \leq n} cov(X_i, X_j) \end{aligned}$$

- ▶ 特别的,

$$D(X_i + X_j) = D(X_i) + D(X_j) + 2cov(X_i, X_j)$$

协方差矩阵的性质: $cov(X_i, X_j) = E[(X_i - EX_i)(X_j - EX_j)]$



- ▶ 对称性: $cov(X_i, X_j) = cov(X_j, X_i)$, 由定义可直接推得, 从而协方差阵 B 是对称矩阵;
- ▶ 协方差矩阵 B 是非负定矩阵, 事实上, 对任意向量 $y = (y_1, y_2, \dots, y_n)$,

$$\begin{aligned} yBy' &= \sum_{i,j} y_i y_j b_{ij} = \sum_{i,j} y_i y_j E[(X_i - EX_i)(X_j - EX_j)] \\ &= E \sum_{i,j} y_i (X_i - EX_i) \cdot y_j (X_j - EX_j) \\ &= E \left[\sum_i y_i (X_i - EX_i) \right]^2 \geq 0 \end{aligned}$$

例 0.1 设二维随机变量 (X, Y) 的联合密度为

$$p(x, y) = \begin{cases} 3x, & 0 < y < x < 1, \\ 0, & \text{其他} \end{cases}$$

试求 $\text{cov}(X, Y)$.

解: 由 $\text{cov}(X, Y) = E(XY) - EXEY$ 知, 我们只需计算 $E(XY), EX, EY$ 的值.

$$\begin{aligned} EX &= \int_{-\infty}^{+\infty} xp_X(x)dx = \int_{-\infty}^{+\infty} x \int_{-\infty}^{+\infty} p(x, y)dydx \\ &= \int_0^1 x \int_0^x 3xdydx = \int_0^1 3x^3dx = 3/4 \end{aligned}$$

$$EY = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} yp(x, y)dydx = \int_0^1 \int_0^x 3xydydx = 3/8$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xyp(x, y)dydx = \int_0^1 \int_0^x xy \cdot 3xdydx = 3/10$$

$$\text{cov}(X, Y) = \frac{3}{10} - \frac{3}{4} \cdot \frac{3}{8} = \frac{3}{160} > 0$$

例 0.2 设二维随机变量 (X, Y) 的联合密度为

$$p(x, y) = \begin{cases} \frac{x+y}{3}, & 0 < x < 1, 0 < y < 2, \\ 0, & \text{其他} \end{cases}$$

试求 $D(2X - 3Y + 8)$.

解: 注意到

$$\begin{aligned} D(2X - 3Y + 8) &= D(2X - 3Y) = D(2X) + D(-3Y) + 2\text{cov}(2X, -3Y) \\ &= 2^2 D(X) + (-3)^2 D(Y) + 2 \cdot 2 \cdot (-3) \text{cov}(X, Y) \\ &= 4D(X) + 9D(Y) - 12\text{cov}(X, Y) \end{aligned}$$

所以我们需要计算: $E(X), E(X^2), E(Y), E(Y^2), E(XY)$.

$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xp(x, y)dydx = \int_0^1 \int_0^2 x \frac{x+y}{3} dydx = 5/9$$

$$E(X^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 p(x, y)dydx = \int_0^1 \int_0^2 x^2 \frac{x+y}{3} dydx = 7/18$$

类似的可计算

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} yp(x,y)dydx = \int_0^1 \int_0^2 y \frac{x+y}{3} dydx = 11/9$$

$$E(Y^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y^2 p(x,y)dydx = \int_0^1 \int_0^2 y^2 \frac{x+y}{3} dydx = 16/9$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xyp(x,y)dydx = \int_0^1 \int_0^2 xy \frac{x+y}{3} dydx = 2/3$$

从而

$$D(X) = 7/18 - (5/9)^2 = 13/162,$$

$$D(Y) = 16/9 - (11/9)^2 = 23/81,$$

$$\text{cov}(X, Y) = 2/3 - 5/9 \cdot 11/9 = -1/81$$

$$D(2X - 3Y + 8) = 4 \cdot 13/162 + 9 \cdot 23/81 - 12 \cdot (-1/81) = 245/81$$

二维正态分布的协方差矩阵



若 $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ 即其联合分布密度为

$$p(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \times \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right] \right\}$$

则由上一章的知识知: $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, 故

$$E(X) = \mu_1, \quad E(Y) = \mu_2$$

$$b_{11} = \text{cov}(X, X) = D(X) = \sigma_1^2,$$

$$b_{22} = \text{cov}(Y, Y) = D(Y) = \sigma_2^2$$

$$\begin{aligned}
b_{12} &= \text{cov}(X, Y) = E[(X - EX)(Y - EY)] = \iint (x - \mu_1)(y - \mu_2)p(x, y)dx dy \\
&\stackrel{\substack{u=(x-\mu_1)/\sigma_1 \\ v=(y-\mu_2)/\sigma_2}}{=} \iint \frac{\sigma_1\sigma_2 uv}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(u^2 - 2\rho uv + v^2)\right\} dudv \\
&= \iint \frac{\sigma_1\sigma_2 uv}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}[(u - \rho v)^2 + (1 - \rho^2)v^2]\right\} dudv \\
&\stackrel{\substack{s=(u-\rho v)/\sqrt{1-\rho^2} \\ t=v}}{=} \frac{\sigma_1\sigma_2}{2\pi} \iint (\sqrt{1-\rho^2}st + \rho t^2) \exp\left\{-\frac{s^2 + t^2}{2}\right\} ds dt \\
&= \frac{\sigma_1\sigma_2}{2\pi} \left(\sqrt{1-\rho^2} \int_{-\infty}^{+\infty} s e^{-\frac{s^2}{2}} ds \int_{-\infty}^{+\infty} t e^{-\frac{t^2}{2}} dt + \rho \int_{-\infty}^{+\infty} e^{-\frac{s^2}{2}} ds \int_{-\infty}^{+\infty} t^2 e^{-\frac{t^2}{2}} dt \right) \\
&= \frac{\sigma_1\sigma_2}{2\pi} (\sqrt{1-\rho^2} \cdot 0 + \rho \sqrt{2\pi} \cdot \sqrt{2\pi}) = \rho\sigma_1\sigma_2
\end{aligned}$$

故二维正态分布的协方差矩阵为

$$B = \begin{pmatrix} \sigma_1^2, & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2, & \sigma_2^2 \end{pmatrix}$$

定义 0.3 设 X, Y 为方差存在的两个随机变量, 则称

$$r := \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}}$$

为 X 与 Y 的相关系数.

相关系数的性质: $r := \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}}$



- ▶ $r = 0 \Leftrightarrow \text{cov}(X, Y) = 0$ 即 X, Y 不相关;
- ▶ 若记 $X^* := \frac{X - EX}{\sqrt{D(X)}}$, $Y^* := \frac{Y - EY}{\sqrt{D(Y)}}$, 则 $EX^* = EY^* = 0$, 从而

$$\begin{aligned} r &:= \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}} = \frac{E[(X - EX)(Y - EY)]}{\sqrt{D(X)}\sqrt{D(Y)}} \\ &= E(X^* Y^*) = E[(X^* - EX^*)(Y^* - EY^*)] \\ &= \text{cov}(X^*, Y^*) \end{aligned}$$

- ▶ $|r| \leq 1$, 且
 - ▶ $r = 1$ 当且仅当 $P(X^* = Y^*) = 1$;
 - ▶ $r = -1$ 当且仅当 $P(X^* = -Y^*) = 1$.

相关系数绝对值小于等于 1 的证明



▶ 首先由 $X^* = \frac{X - EX}{\sqrt{D(X)}}$, $Y^* = \frac{Y - EY}{\sqrt{D(Y)}}$ 知:

$$EX^* = EY^* = 0$$

$$D(X^*) = D\left(\frac{X - EX}{\sqrt{D(X)}}\right) = \frac{1}{D(X)}D(X - EX) = \frac{1}{D(X)}D(X) = 1$$

$$D(Y^*) = D\left(\frac{Y - EY}{\sqrt{D(Y)}}\right) = \frac{1}{D(Y)}D(Y - EY) = \frac{1}{D(Y)}D(Y) = 1$$

▶ 其次, 由 $r = \text{cov}(X^*, Y^*)$ 知

$$\begin{aligned}|r| &= |\text{cov}(X^*, Y^*)| = |E(X^* Y^*) - E(X^*)E(Y^*)| \leq \sqrt{E[(X^*)^2]E[(Y^*)^2]} \\ &= \sqrt{D(X^*)D(Y^*)} = 1;\end{aligned}$$

▶ $|r| = 1$ 当且仅当存在 t_0 使得 $P(Y^* = t_0 X^*) = 1$, 而此时有

$$r = E(X^* Y^*) = t_0 E[(X^*)^2] = t_0$$

随机变量不相关时期望与方差的性质



定理 0.1 对于随机变量 X, Y , 下面四个事实是等价的:

1. X, Y 不相关;
2. $r = 0$ 即相关系数为 0;
3. $E(XY) = E(X)E(Y)$;
4. $D(X+Y) = D(X) + D(Y)$.

证明: (1) 与 (2) 等价是显然的, 根据定义即可得. 由于

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y)$$

故 $\text{cov}(X, Y) = 0$ 当且仅当 $E(XY) = E(X)E(Y)$, 即 (1) 和 (3) 等价. 另外, 又由于

$$D(X+Y) = D(X) + D(Y) + 2\text{cov}(X, Y)$$

故 $\text{cov}(X, Y) = 0$ 当且仅当 $D(X+Y) = D(X) + D(Y)$, 即 (1) 与 (4) 等价.

独立与不相关的联系：独立必定不相关



定理 0.2 若 X, Y 独立, 则 X 与 Y 不相关.

证明: 我们仅对连续型随机变量给出证明. 因为 X, Y 独立, 故其联合分布密度 $p(x, y) = p_X(x)p_Y(y)$, 从而

$$\begin{aligned} E(XY) &= \iint xyp(x, y)dxdy = \iint xyp_X(x)p_Y(y)dxdy \\ &= \int xp_X(x)dx \int yp_Y(y)dy = E(X)E(Y) \end{aligned}$$

从而 X, Y 不相关.

定理 0.3 若 X, Y 独立, 则

$$E(XY) = E(X)E(Y), \quad D(X+Y) = D(X) + D(Y)$$

更一般的, 我们有若 $X_1, \dots, X_n$ 为相互独立的随机变量, 则.

$$E(X_1 \cdots X_n) = E(X_1)E(X_2) \cdots E(X_n);$$

$$D\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n D(X_i)$$



例 0.3 设随机变量 $X \sim N(0, \sigma^2)$, 并令 $Y = X^2$, 显然 X, Y 不独立, 但是此时 X, Y 不相关。事实上, 此时

$$\text{cov}(X, Y) = E(X \cdot X^2) - E(X)E(X^2) = 0.$$

例 0.4 设 θ 为 $[0, 2\pi]$ 上的均匀分布, a 为一固定常数, 令

$$X = \cos \theta, \quad Y = \cos(\theta + a).$$

则我们有

$$\begin{aligned} E(X) &= \int_0^{2\pi} \cos t \cdot \frac{1}{2\pi} dt = 0, & E(Y) &= \int_0^{2\pi} \cos(t+a) \cdot \frac{1}{2\pi} dt = 0 \\ E(X^2) &= \int_0^{2\pi} \cos^2 t \cdot \frac{1}{2\pi} dt = \frac{1}{2}, & E(Y^2) &= \int_0^{2\pi} \cos^2(t+a) \cdot \frac{1}{2\pi} dt = \frac{1}{2} \\ E(XY) &= \int_0^{2\pi} \cos t \cos(t+a) \cdot \frac{1}{2\pi} dt = \frac{1}{2} \cos a \end{aligned}$$

因此 $r = \cos a$. 故

- ▶ $a = 0$ 时, $r = 1, X = Y$;
- ▶ $a = \pi$ 时, $r = -1, X = -Y$;
- ▶ $a = \frac{\pi}{2}$ 或 $\frac{3\pi}{2}$ 时, $r = 0, X, Y$ 不相关, 但此时 $X^2 + Y^2 = 1$, 因此不独立.

二维正态分布：不相关与独立等价



定理 0.4 对于二维正态分布 (X, Y) , X 与 Y 不相关与 X, Y 独立等价.

证明: 因为独立必然不相关, 故我们仅需证在不相关条件下, X, Y 独立即可. 对于二维正态分布, 我们有

$$\text{cov}(X, Y) = \rho\sigma_1\sigma_2$$

从而 X, Y 不相关, 当且仅当 $\rho = 0$, 此时

$$\begin{aligned} p(x, y) &= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \times \\ &\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right\} \\ &= \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left\{-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right\} \cdot \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left\{-\frac{(y-\mu_2)^2}{2\sigma_2^2}\right\} \\ &= p_X(x)p_Y(y) \end{aligned}$$

故 X, Y 独立, 定理得证.